Quantitative Portfolio Management Final Project

High-Quality Earnings Momentum

Exploiting Underreaction in Profitable, Conservative-Growth Firms

Group 12

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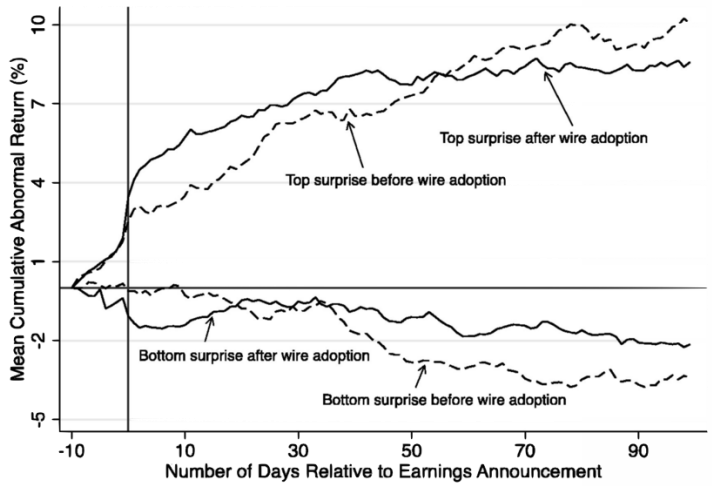
Motivation: Why Should Alpha Exist?



SUE

×

High Profitability

×

Low Asset Growth

Sticky Belief Revision

Fundamental news is incorporated into prices gradually, not immediately.

↓

Persistent Profitability

Profitable firms tend to maintain strong earnings; market underestimates the durability of earnings shocks

↓

Conditional Underreaction

Underreaction is strongest in firms with high profitability and conservative growth

↓

The strategy bets that the market systematically underreacts to durable, high-quality earnings — and that conditioning on investment discipline concentrates this effect.

Connection to the Literature

Phenomenon	*Key Paper(s)*	Our Connection
Post-earnings announcement drift (PEAD)	*Ball & Brown (1968); Bernard & Thomas (1989); Foster et al. (1984)*	SUE as primary signal; captures gradual price adjustment to fundamental news.
Gross profitability premium	*Novy-Marx (2013)*	GP filter isolates high-quality earnings with persistence, reducing noise.
Asset growth anomaly	*Cooper, Gulen & Schill (2008)*	AG filter removes firms engaging in aggressive investment, improving signal precision by mitigating over-extrapolation.

$$\mathcal{U}_t = \{ i : \; GP_{i,t-1} \geq P_{70}(GP), \quad AG_{i,t-1} \geq P_{30} \}$$

$$HQEM_t = R_t(Q_{\mathrm{SUE}} \mid i \in \mathcal{U}_t) - R_t(Q1_{\mathrm{SUE}} \mid i \in \mathcal{U}_t)),$$

weights $\geq$ lagged market cap

We integrated three well-documented return predictors into a single, theoretically coherent framework and test whether their interaction generates incremental alpha.

Strategy Construction: Hierarchical Filtering + SUE Ranking

1 Universe Selection

NYSE, AMEX, NASDAQ stocks with sufficient accounting history

2 Quality Filter (Gross Profitability)

Retain firms with GP ≥ 70th NYSE percentile (High Quality)

3 Investment Filter (Asset Growth)

Keep firms with AG ≤ 30th NYSE percentile (Conservative Growth)

4 SUE Ranking & Portfolio Formation

Long top 20% (Q5), Short bottom 20% (Q1) based on SUE

Technical Specifications

Rebalancing

Monthly rebalancing; Dollar-neutral (~119 long / ~113 short stocks, full-sample average)

Weighting Scheme

Value-weighted by lagged market cap

Data Processing

AG winsorized at 1%/99%; NYSE breakpoints used to prevent small-cap bias

The funnel concentrates the earnings surprise signal in the corner of the universe where it is theoretically most powerful.

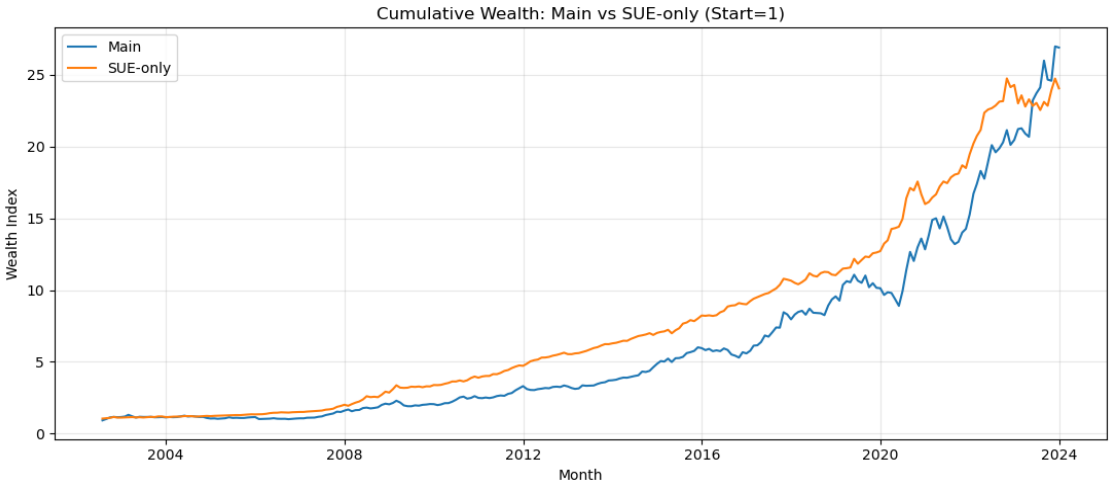
$$HQEM_t = R_t\big(\text{High SUE} \mid GP_{t-1} \geq P70,\ AG_{t-1} \leq P30\big) - R_t\big(\text{Low SUE} \mid GP_{t-1} \geq P70,\ AG_{t-1} \leq P30\big)$$

Main Results

Metric	Main Strategy	SUE-Only	GP-Only	AG-Only
Ann. Mean Return	**18.03%**	14.38%	3.93%	2.07%
Ann. Volatility	16.04%	7.41%	10.08%	10.04%
Sharpe Ratio	0.97	**1.58**	0.14	-0.03
Max Drawdown	−22.58%	-8.07%	-30.5%	-32.4%
Monthly Mean (t-stat)	1.50% (6.68)	1.2%（11.26）	0.33%（2.06）	0.17%（1.09）

Sample: WRDS_1990 dataset, Jan 1990 – Dec 2023. Benchmark: Risk-free rate.

- **Higher return, lower efficiency:** Main strategy earns 18.0% annually vs 14.4% for SUE-only, but Sharpe falls from 1.58 to 0.97.
- **Tail risk remains meaningful:** Max drawdown is -22.6%, so the filters improve the economic story, not eliminate downside risk.
- **Raw performance is not alpha yet:** These are unconditional returns; the next slide tests whether performance survives standard factor controls.



The main strategy has higher raw return than SUE-only, but worse risk-adjusted efficiency.

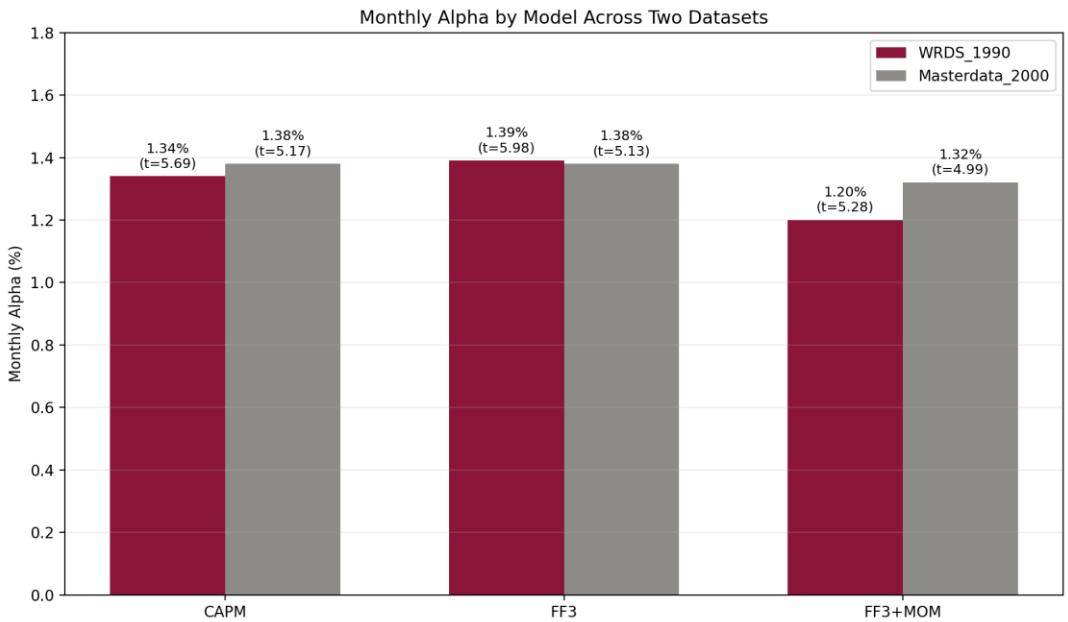
Alpha as a Pricing Test: Risk-Adjusted Returns

WRDS_1990 Sample (1990–2023)

Model	Monthly α	t(α)	β_{mkt}	β_{mom}	R^2
CAPM	1.34%	5.69	–0.055	—	0.003
FF3	1.39%	5.98	–0.103	—	0.044
FF3+MOM	**1.20%**	**5.28**	–0.009	0.269	0.109

Masterdata_2000 Sample (2002.07–2023)

Model	Monthly α	t(α)	β_{mkt}	β_{mom}	R^2
CAPM	1.38%	5.17	—	—	—
FF3	1.38%	5.13	—	—	—
FF3+MOM	**1.32%**	**4.99**	—	0.218	0.069



Alpha persists across models: Monthly alpha remains around 1.2%–1.4% with strong t-stats across CAPM, FF3, and FF3+MOM.

Cross-dataset consistency: The result is replicated in both the WRDS and Masterdata samples, reducing concern that it is driven by one cleaning pipeline.

Some momentum exposure, but not the whole story: The strategy has positive momentum beta, yet substantial alpha remains after controlling for it.

Main caveat: Because the strategy is built from profitability and investment filters, FF5-style controls are still needed to rule out omitted RMW/CMA exposure.

Alpha remains statistically significant after standard factor controls in both datasets.

Robustness Tests

WRDS_1990 (Main Sample)

Test Specification	Ann. Ret	Sharpe	α (FF3+MOM)	t(α)
Full Sample (1990–2023)	18.03%	0.97	1.20%	5.28
Excl. Bottom 20% ME	14.67%	0.77	0.98%	4.31
Subperiod: 1990–1999	15.93%	0.81	—	—
Subperiod: 2000–2023	18.82%	1.03	—	—
Subperiod: 2005–2023	17.41%	1.14	—	—

Masterdata_2000 (Cross-check)

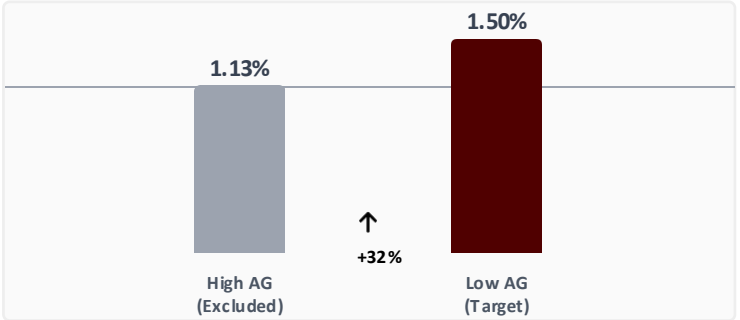
Test Specification	Ann. Ret	Sharpe	α (FF3+MOM)	t(α)
Full Sample (2002–2023)	16.49%	1.02	1.32%	4.99
Excl. Bottom 20% ME	14.45%	0.90	1.07%	4.11
Subperiod: 2002–2012	12.38%	0.76	—	—
Subperiod: 2013–2023	20.41%	1.26	—	—
(Sample constrained by dataset availability)				

- **Microcap Exclusion:** Alpha declines by ~0.20% per month when excluding the bottom 20% of stocks, but remains economically large (~1.0%) and statistically significant ($t > 4.0$).
- **Temporal Stability:** Risk-adjusted performance improves in later subperiods, as reflected in higher Sharpe ratios after 2005 and after 2013.
- **Dataset Consistency:** Results are robust across two independent data preparations, confirming the signal is not an artifact of specific cleaning choices.
- **Short Leg Dependence:** While robust, the strategy relies on shorting small/mid-cap growth stocks, raising potential implementation cost concerns.

Performance is robust across datasets and samples, but small-cap exposure and short-leg implementability remain practical concerns.

Does Asset Growth Filter Sharpen the SUE Signal?

Subsample Group	Avg. Monthly SUE H-L Return
High GP / Low AG (Target Strategy)	1.50%
High GP / High AG (Excluded "Empire Builders")	1.13%
Performance Lift	**+0.37%**



- **Economic Confirmation:** The SUE high-minus-low spread is about **32% stronger** in the **Low-AG** subset than in the **High-AG** subset within high-profitability firms.
- **Mechanism:** One possible interpretation is that high-AG firms may be more prone to over-investment, so positive earnings surprises are less persistent and the SUE signal becomes weaker.
- **Interaction Effect:** This is not merely factor exposure; the *interaction* of Gross Profitability × Low Asset Growth × SUE creates the alpha.
- **Uncertainty Channel:** Low-AG firms may have more predictable earnings structures, reducing fundamental uncertainty and allowing cleaner price adjustment trends (PEAD).

? ***Skeptical Check:*** Is the roughly **0.36%–0.37%** monthly differential structural, or could part of it reflect estimation noise? Further testing is needed to assess the stability of this mechanism.

The data supports the proposed mechanism: Asset Growth conditioning strengthens the earnings surprise signal, consistent with underreaction in high-quality, capital-disciplined firms.

Conclusion and Open Questions

Summary Findings

- **Robust Alpha:** The long-short strategy (High GP + Low AG + SUE) delivers **1.20%–1.32% monthly alpha (FF3+MOM)** across **two independent data constructions**, with samples spanning **1990–2023** in WRDS and **2002–2023** in the cross-check dataset.
- **Economic Channel:** The evidence is **consistent with** the idea that conditioning on investment discipline strengthens the PEAD effect.
- **Size Robustness:** Alpha survives microcap exclusion with t-statistics > 4 in all robustness samples.
- **Stability**: The strategy remains market-neutral and value-weighted, with positive performance across subsamples and stronger Sharpe ratios in later periods.

Open Questions & Next Steps

- **Factor Exposure:** FF5 test is critical; Profitability (RMW) and Investment (CMA) factors might explain the alpha.
- **Implementation:** What fraction of return comes from the short leg? Are these positions borrowable at scale?
- **Transaction Costs:** Despite monthly rebalancing, turnover in the short leg may erode net returns.
- **Regime Analysis:** Why does performance improve post-2005? Is this due to liquidity changes or factor behavior?
- **Data Mining:** True out-of-sample testing (e.g., international markets) is needed to rule out overfitting.

Core Interpretation: Alpha is likely a mix of risk compensation (omitted factors like CMA), mispricing (behavioral underreaction), and possible data mining — disentangling these requires additional tests.

A compelling strategy with strong empirical support — but intellectually honest about its limitations.